Spin, No-Cloning, and Distributivity Failure

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Abstract

This paper is a brief pedagogical introduction to spin, with a discussion of the logical constraints on our observables. This includes a brief proof of the No-Cloning theorem along with a proof of the failure of distributivity.

1 Introduction

In 1922, Stern and Gerlach demonstrated an intrinsic property of silver. We would go on to realize this was, in fact, a general property of particles known as *spin*.

The Stern-Gerlach experiment sent a beam of silver atoms through a magnetic field. When this beam passed through the field, the atoms were either deflected North $(+z)$ or South $(-z)$ with a 50/50 split. If we again project the $(+z)$ Ag atoms through a magnetic field, they all deflect $(+z)$. Likewise, $(-z)$ continue deflecting $(-z)$. However, if the $(+z)$ atoms pass through a magnetic field in the $(+x)$ direction, they deflect $(+x)$ 50% of the time and $(-x)$ 50% of the time. Furthermore, if we then deflect our $(+x)$ atoms back through $(+z)$, the split is once again 50%/50% [1]

We may observe from these reproducible experiments there is an intrinsic property in particles, known as spin, which we can represent the state of as a vector in a Hilbert space. Furthermore, we can define it's measurement and the possible resulting states.

1.1 Axioms

Definition 1.1 (Inner Product Space)**.** Let V be a vector space over the field $\mathbb{F}$. Then V is an inner product space if we have a function $\langle\cdot|\cdot\rangle : V \times V \to \mathbb{F}$ that obeys four properties:

1. **Conjugate Symmetry**: $\forall\phi \in V, \forall\psi \in V, \langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$
2. **Positive Definite**: $\forall\phi \in V, \langle\phi|\phi\rangle \geq 0$
3. **Definite**: $\forall\phi \in V, \langle\phi|\phi\rangle = 0 \implies \phi = 0$
4. **Bilinear**: $\forall\phi, \psi, \omega \in V, \forall a \in \mathbb{F}, \langle a\phi + \psi|\omega\rangle = a\langle\phi|\omega\rangle + \langle\psi|\omega\rangle$

We will assume our vector space is over the field $\mathbb{C}$ from now on.

Definition 1.2 (Complete)**.** A vector space V is complete iff for every Cauchy sequence $x_n \in V$, $\exists x \in V, \lim_{n\to\infty}(x_n) = x$.[1]

Definition 1.3 (Hilbert Space)**.** A Hilbert space $\mathcal{H}$ is a complete inner-product space.

Definition 1.4 (Hermitian)**.** A is Hermitian iff $A = A^\dagger$.[2]

Now we will define our postulates [2]

Postulate 1. *The state of a quantum mechanical system is completely specified by a complex vector $|\psi\rangle$ in a Hilbert space $\mathcal{H}$.*

[1] A Cauchy sequence is mathematically equivalent to a convergent sequence in a complete space.
[2] $\dagger$ is conjugate transpose.

Postulate 2. *Every observable is represented by a Hermitian operator A acting on $\mathcal{H}$.*

Postulate 3. *Every resultant measurement is an eigenvalue λ_n of the corresponding operator A.*

Postulate 4. *The probability of measuring eigenvalue λ_n is given by $\mathcal{P}(\lambda_n) = |\langle\lambda_n|\psi\rangle|^2$.*

Postulate 5. *The post measurement state is the normalized projection onto the eigenvector corresponding to λ_n.*[3]

The force $\mathbf{F}$ experienced by the atoms in the experiment's magnetic field $\mathbf{B}$ is:

$$\mathbf{F_z} \propto S_z \frac{\partial B_z}{\partial z}\hat{\mathbf{k}} \tag{1}$$

Where $\mathbf{S}$ is our Hermitian observable known as spin. The state of our particle pre-measurement is $|\psi\rangle$. Afterward the state has been projected onto (+z) or (-z).

1.2 Notes on our Postulates

The Stern-Gerlach experiment we discussed specifically analyzed a spin-1/2 system. This means our Hilbert space has a dimension of 2. We will not delve into any other system here.

I encourage those reading this for the first time to think of our observables as primitives that relate the geometry of our Hilbert space to the possible measurements we can make on a particle's spin.

[3] We will omit our last postulate for the sake of brevity: our time evolution obeys Schrödinger's equation.

1.3 Standard Basis

We will denote our standard basis vectors of our Hilbert space $\mathcal{H}$ as $|1\rangle$ and $|0\rangle$. These are the wave-states for particles sent in the (+z) and (-z) directions respectively. We can represent our basis vectors and Pauli matrices (σ_i) as follows[4]:

$$|1\rangle \doteq \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |0\rangle \doteq \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \sigma_z \doteq \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{2}$$

We will define the wave-states for the x and y spacial directions in terms of representations of our basis vectors:

$$|1\rangle_x \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, |0\rangle_x \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \sigma_x \doteq \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{3}$$

$$|1\rangle_y \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, |0\rangle_y \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}, \sigma_y \doteq \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{4}$$

$\sigma_x, \sigma_y, \sigma_z,$ and $\mathbb{I}$ define our complete basis for our Hermitian matrices.

Example 1.1. $Herm(2) = \text{span}\{\sigma_x, \sigma_y, \sigma_z, \mathbb{I}\}$

We define our special orthogonal two group as follows[5]:

$$\text{SU}(2) \doteq \{U : U \in GL(2, \mathbb{C}) \wedge det(U) = 1\} \tag{5}$$

This matrix group forms all the operators that spatially rotate our vectors. This is not the same as the Hermitian operators. Our Hermitian matrices are always observables, whilst our unitary matrices are what evolve our system.

[4] $\doteq$ means "represented by". Technically our operators are group actions on our Hilbert space.

[5] GL(2, $\mathbb{C}$ means the group of invertible 2 by 2 matrices with complex values (it stands for general linear).

1.4 Projection Operators

Definition 1.5 (Projection Operator)**.** An operator is a projection $P : \mathcal{H} \to \mathcal{H}$ iff $P^2 = P$.

We can project a wave state onto $|\psi\rangle$ with the outer-product $|\psi\rangle\langle\psi|$.

2 Failure of Distributivity

Let $\mathcal{H} \supset A := \text{span}\{|1\rangle_z\}$.

Let $\mathcal{H} \supset B := \text{span}\{|1\rangle_x\}$.

Let $\mathcal{H} \supset C := \text{span}\{|0\rangle_x\}$

Theorem 1. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Proof. $\emptyset = \emptyset$ □

Theorem 2. $A \cap \text{span}\{B \cup C\} \neq (A \cap B) \cup (A \cap C)$.

Proof. $\text{span}\{B \cup C\}$ generates $\mathcal{H}$. Thus $A \cap \text{span}\{B \cup C\} = A$. $A \neq \emptyset$. □

To quote Aristotle, "The whole is greater than the part." The theme here is we must account for superposition as a physical reality. We cannot say the particle is in (+x) or disjointly in (-x) when it is in superposition.

Quantum theorists defined a proposition as a certain kind of Hilbert space[6]. Assuming as much, you can observe non-distributivity in Thm. 2. However, this does not violate classical mechanics; it should be judiciously noted that $\text{span}\{B \cup C\}$ is the pre-measurement subspace, whilst B and C in isolation

[6] A closed subspace

are both post measurement subspaces; this warrants careful treatment when formulating propositional logic in quantum mechanics.

3 Tensors

Definition 3.1 (Tensor Product)**.** A tensor product is a bilinear mapping[7] that satisfies the universal property.

$$\otimes : \mathcal{H} \times \mathcal{H} \to \mathcal{H} \otimes \mathcal{H} \tag{6}$$

Bilinearity: $\forall \psi, \phi, \omega, \in \mathcal{H}, a \in \mathbb{C}$

1. $a\psi \otimes \phi = a(\psi \otimes \phi) = \psi \otimes a\phi$
2. $(\psi + \phi) \otimes \omega = (\psi \otimes \omega) + (\phi \otimes \omega)$
3. $\psi \otimes (\phi + \omega) = (\psi \otimes \phi) + (\psi \otimes \omega)$

Associativity: $(\phi \otimes \psi) \otimes \omega = \phi \otimes (\psi \otimes \omega)$

[8] This mapping gives the mathematics for systems of multiple particles which we empirically observe to be bilinear and associative[9]. For the pedantic reader, technically associativity only holds up to natural isomorphism, but the average user can treat tensor products as in Def. 3.1. Additionally, operators acting on the new Hilbert space $\mathcal{H} \otimes \mathcal{H}$ are the tensored vector spaces of our represented operators:

$$\forall U, V \in \mathcal{M}(2, \mathbb{C}) \forall \psi, \phi \in \mathcal{H}, (U \otimes V)(\psi \otimes \phi) = (U\psi) \otimes (V\phi). \tag{7}$$

[7]Think of a tensor product as a function $\otimes(\psi, \phi)$ rather than an operator if this helps (these two things are technically the same).

[8]In general our tensor product is not necessarily between the Hilbert space $\mathcal{H}$ and itself, but we like to use 1/2 systems (dimension 2) systems in quantum computing and this is what I'll stick to.

[9]Associativity holds in our tensor definition by the universal property.

Notice our tensored Hilbert space has a dimension of 4. Our basis vectors form the set $\{|0\rangle \otimes |0\rangle, \{|0\rangle \otimes |1\rangle, \{|1\rangle \otimes |0\rangle, \{|1\rangle \otimes |1\rangle\}$

Definition 3.2 (Composite System)**.** Wave states $|\psi\rangle$ and $|\phi\rangle$ form a composite system $|\psi\rangle \otimes |\phi\rangle$

4 No-Cloning Theorem

In the next theorem, assume all wave-states are normalized.

Theorem 3 (No-Cloning Theorem)**.**

$$\nexists U \in U(4) \forall \phi, \psi \in \mathcal{H}, U(\phi \otimes \psi) = (\phi \otimes \phi) \tag{8}$$

Proof. Suppose for the sake of contradiction that for states $\phi, \omega \in \mathcal{H}$:

$$U(|\phi\rangle \otimes |\psi\rangle) = (|\phi\rangle \otimes |\phi\rangle) \wedge U(|\omega\rangle \otimes |\psi\rangle) = (|\omega\rangle \otimes |\omega\rangle)$$

Then

$$(\langle\phi| \otimes \langle\psi|)(|\omega\rangle \otimes |\psi\rangle) = (\langle\phi| \otimes \langle\psi|)U^\dagger U(|\omega\rangle \otimes |\psi\rangle) = e^{i\theta}(\langle\phi| \otimes \langle\phi|)(|\omega\rangle \otimes |\omega\rangle) = e^{i\theta}\langle\phi|\omega\rangle^2$$

But we also have:

$$\langle\phi|\omega\rangle = (\langle\phi| \otimes \langle\psi|)(|\omega\rangle \otimes |\psi\rangle)$$

Thus $\langle\phi|\omega\rangle = e^{i\theta}\langle\phi|\omega\rangle^2 \implies |\langle\phi|\omega\rangle| = |\langle\phi|\omega\rangle|^2$. The only values that satisfy this equation are 0 or 1.[10] Thus $|\phi\rangle = e^{i\theta}|\omega\rangle$ (invariant up to a phase factor) or $|\phi\rangle$ is mutually orthogonal to $|\omega\rangle$ [3]. □

[10] Postulate 4 is implicitly used here. Our inner-product must be positive, real, and a probability

References

[1] Allan Franklin and Slobodan Perovic. *Experiment in Physics ¿ Appendix 5: Right Experiment, Wrong Theory: The Stern-Gerlach Experiment (Stanford Encyclopedia of Philosophy)*. plato.stanford.edu, 2023. URL: `https://plato.stanford.edu/entries/physics-experiment/app5.html`.

[2] David H McIntyre. *Quantum Mechanics*. Cambridge University Press, Sept. 2022.

[3] *no-cloning theorem in nLab*. ncatlab.org, Aug. 2023. URL: `https://ncatlab.org/nlab/show/no-cloning+theorem`.